Proposed recovery integration contract

The reducer is pure. It does not implement a durable worker, real SIS connector or cryptographic verifier.

Worker: obtain tenant-scoped job lease; load command; validate observation against command, attempt and request digest; call reducer; compare-and-swap expected version and lease token; append audit and outbox effects atomically. Execute effects only through allowlisted workers.

Expired executing lease is ambiguous delivery: emit TIMEOUT, not a fresh write. Persist next_check_at, query budget, deadline and stable operation key. Bound status-query backoff; observe Retry-After. No network call inside database transaction.

Every fresh dispatch must recheck authority, approval binding, epoch, kill switch and business preconditions. The reducer never sends a write. Late successful acknowledgment records a completed historical action even if current write permission is revoked.

NOT_APPLIED permits reevaluation only if original request cannot later complete. NOT_FOUND or eventual-consistency absence is insufficient. Query-budget exhaustion requires manual review, not silent failure.

Audit format is illustrative. Protect append paths, tenant reads, key registry and independently retained signed checkpoints. No secrets or raw student record contents in normal logs.

Read-only repository search at commit 57185b085d88ad6bf510b2a57ecfea3d97d99292 documents real SIS connectors as PLANNED/BLOCKED and examples as fake in-memory SIS. No external state changed.